The Structure of Local Motion: Unpacking the State Transition Matrix

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Abstract

The local motion about a trajectory in a Hamiltonian Dynamical system, in particular in an astrodynamics system, is controlled by the state transition matrix. By analyzing this solution one can obtain insights into the local stability and instability of motion, can identify constraints on the volumetric propagation of solutions, and other key insights. In this talk we will introduce and discuss some of the fundamental structures that can be found in the state transition matrix and introduce a novel approach to analyzing general trajectories. Examples from the 2-body problem are given.

1 Introduction

A key component for analyzing motion in astrodynamics problem is the state transition matrix (STM), which describes how the local phase space maps in the vicinity of a given trajectory. There are a plethora of applications for the STM, including stability and instability analysis, computation of periodic and quasi-periodic motions, use in mapping uncertainty distributions, and determining sensitivity of a trajectory to state or parameter errors. In this note we analyze the fundamental structure associated with the STM in terms of the Hamiltonian system parameters from which it is derived. In particular, we introduce a recently derived ortho-symplectic transformation that can be applied to separate the basic flow components associated with an STM, enabling the definition of a local Poincaré-like map that can be defined between any two points on a trajectory, and which devolves to the classical Poincaré map and associated monodromy matrix when evaluated over a periodic motion.

2 Hamiltonian Dynamical System Basics

Recall that a Hamiltonian Dynamical system is described as

$$\dot{\mathbf{x}} = JH_{\mathbf{x}}(\mathbf{x})$$

where $H(\mathbf{x})$ is the Hamiltonian function, $\mathbf{x} = [\mathbf{q},\ \mathbf{p}] \in \mathbb{R}^{2n\times 2n}$ and

$$J = \begin{bmatrix} \mathbf{0}_{n\times n} & \mathbf{I}_{n\times n} \\ -\mathbf{I}_{n\times n} & \mathbf{0}_{n\times n} \end{bmatrix}$$

Given a general state $\mathbf{x}_o$ and a time t_o, the resulting flow of the dynamical system driven by the Hamiltonian Dynamical System (HDS) can be described as $\mathbf{x}(t) = \phi(t; \mathbf{x}_o, t_o)$ and satisfies the HDS: $\partial\phi/\partial t = J\ \partial H/\partial\mathbf{x}|_{\phi}$.

The STM is defined as the partial of the solution flow with respect to the initial condition $\mathbf{x}_o$, $\Phi(t, t_o, \mathbf{x}_o) = \frac{\partial\phi}{\partial\mathbf{x}_o}$. The STM also satisfies the linearized differential equation

$$\dot{\Phi}(t, t_o) = JH_{\mathbf{xx}}|_{\phi(\mathbf{t})}\Phi(t, t_o)$$

$$\Phi(t_o, t_o) = I$$

3 Local Structure

In [1] a procedure is introduced that derives an ortho-symplectic transformation that is based on the local dynamical flow evaluated at a state, $JH_{\mathbf{x}}$, and the Hamiltonian gradient $H_{\mathbf{x}}$. The procedure starts by defining a basis direction $\mathbf{u}_1 = J\hat{H}_{\mathbf{x}}$ as the first coordinate direction and $\mathbf{v}_1 = -J\mathbf{u}_1 = \hat{H}_{\mathbf{x}}$ as the first momentum direction. The remaining coordinate bases $\mathbf{u}_j$ are constructed to be orthogonal to the previous coordinates $\mathbf{u}_k,\ k < j$ using a Gram-Schmitt type of approach. Then the corresponding momentum bases are defined as $\mathbf{v}_j = -J\mathbf{u}_j$. Grouping the terms together defines the ortho-symplectic transformation that is defined at any state:

$$R(\mathbf{x}) = [\mathbf{u}_1\mathbf{u}_2\cdots\mathbf{u}_n\mathbf{v}_1\mathbf{v}_2\cdots\mathbf{v}_n]$$

In [1] it is shown that this transformation is both symplectic and orthogonal, meaning that $R^T = R^{-1}$ and $RJ = JR$. This transformation can be used to show that phase space in the $\mathbf{u}_1$ and $\mathbf{v}_1$ coordinates, correlated with the fundamental solution flow and gradient of the Hamiltonian at a given state, are uncoupled from the other directions and locally conserve phase volume, implying that the remaining bases also conserve volume separately. In this way the transformation separates the local phase volume into disjoint

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components. In this note we explore the full transformation and develop a concept for using it to simplify and expose the local structure of motion.

The transformation can be defined at any given state, $\mathbf{x}$, and the linear mapping about that state from an initial time t_o, state $\mathbf{x}_o$ and deviation $\delta\mathbf{x}_o$ to a final time t_f, state $\mathbf{x}_f$ and deviation $\delta\mathbf{x}_f$, where $\delta\mathbf{x}_f = \Phi(t_f, t_o)\delta\mathbf{x}_o$. Denote the transformations at the endpoint times as $R(\mathbf{x}_o) = R_o$ and $R(\mathbf{x}_f) = R_f$. Then we can define the transformed state deviations as $\delta\mathbf{X} = R^T\delta\mathbf{x}$. We note that the coordinate $\delta X(1)$ is associated with the solution flow and $\delta X(n+1)$ is associated with the local Hamiltonian gradient. Applying the transformation to the linear mapping changes $\delta\mathbf{x}_f = \Phi\delta\mathbf{x}_o$ into $\delta\mathbf{X}_f = \Psi\delta\mathbf{X}_o$, where $\Psi = R_f^T\Phi R_o$. We note that Ψ is symplectic as the original Φ and the R transformations are all symplectic. The mapping Ψ goes between the basis evaluated at the initial state to the basis evaluated at the final state.

In the new basis, Ψ has some particular structure that is of interest. The first column of Ψ will have a non-zero initial entry, $\Psi_{11} \neq 0$, with all the remaining terms $\Psi_{i1} = 0$ for $i > 1$. This results from the identity $JH_{\mathbf{x}_f} = \Phi JH_{\mathbf{x}_o}$ and represents that offsets along the dynamical flow direction lie along $\delta X(1)$ and do not affect the other states locally. The $n+1$ row of Ψ, associated with the Hamiltonian gradient, has a similar structure with $\Psi_{(n+1)i} = 0$ for $i \neq n+1$ and $\Psi_{(n+1)(n+1)} \neq 0$, representing that the state $\delta X(n+1)$ fully represents the change in the Hamiltonian constant. We also note that the term $\Psi_{1(n+1)}$ is non-zero, and represents the effect of a change in the Hamiltonian constant along the flow direction. We also note that in general the other entries in the first row of Ψ are non-zero, meaning that deviations in them also affect the flow direction.

We can then decouple the STM into two separate sub-matrices. In the following we assume a 2DOF system, so that $n = 2$.

$$\Psi_{13} = \begin{bmatrix} \Psi_{11} & \Psi_{13} \\ 0 & \Psi_{33} \end{bmatrix}$$

As proven in [1], the product $\Psi_{11}\Psi_{33} = 1$, and indicates that the system trades expansion between the flow and Hamiltonian gradient directions. We note that the eigenvalues of this matrix must be real. Similarly, for a 2DOF, we can extract the sub-matrix

$$\Psi_{24} = \begin{bmatrix} \Psi_{22} & \Psi_{24} \\ \Psi_{42} & \Psi_{44} \end{bmatrix}$$

which represents the components of phase space separate from the flow and Hamiltonian gradient directions. This sub-matrix will also have unity determinant, indicating that it is symplectic, however the eigenvalues of this matrix can be either real or on the unit circle. For systems with a higher degree of freedom additional complex cases can arise.

It is significant to point out that the eigenvalues of these sub-matrices are equal to the eigenvalues of the transformed STM Ψ, but are not equal to the eigenvalues of Φ. This arises as the transformation is not a similarity matrix. We note that if the trajectory is in fact periodic, and we evaluate the matrix Ψ after one full period, then the eigenvalues of Ψ and Φ will agree, as then $R_f = R_o$ and the transformation is a similarity transformation.

4 Local Poincaré Map

One use of this transformation is to define a Poincaré map between two arbitrary points along a trajectory. In a standard surface of section approach the monodromy matrix mapping fixes the Hamiltonian value and removes variations along the trajectory. In the reduced map we can enforce the energy conservation by setting $\delta X(3)_o = 0$ and the initial starting surface orthogonal to the trajectory by setting $\delta X(1)_o = 0$. Then the remaining coordinates $\delta X(2)$ and $\delta X(4)$ represent the local orbit variation with the energy and flow direction removed, thus representing a monodromy matrix. We note that variations in these coordinates will make the final surface of section $\delta X(1)_f$ to be non-zero in general. It is possible to map back to the surface by allowing the timing along the orbit to vary, in this way creating a true monodromy matrix. This aspect of the problem will be discussed in more detail at the conference. Finally, if the nominal trajectory is evaluated over a full period, this reduced map represents the associated monodromy matrix.

5 2BP Application

In the presentation this mapping will be evaluated for both the standard 2-Body Problem, and also applied to the Circular Restricted 3-Body Problem. In the current note we just focus on the 2BP, evaluating the motion in a plane. First, recall the Hamiltonian of the 2BP, where we assume normalized coordinates and a unity gravitational parameter:

$$H(\mathbf{q}, \mathbf{p}) = \frac{1}{2}(p_x^2 + p_y^2) - \frac{1}{r}$$

where $r = \sqrt{x^2 + y^2}$.

Although this system is integrable, it makes a useful test case for our approach to deconstructing the STM. Indeed, as stated here the system has two main integrals of motion, the total energy (or Hamiltonian is constant), and an angular momentum integral of the form $h = q_x p_y - q_y p_x$. To start the construction of our special basis, we compute $JH_{\mathbf{x}}$ and $H_{\mathbf{x}}$ to find:

$$\mathbf{u}_1 = JH_{\mathbf{x}} = [p_x, p_y, -x/r^3, -y/r^3]^T$$

$$\mathbf{v}_1 = H_{\mathbf{x}} = [x/r^3, y/r^3, p_x, p_y]^T$$

and note that these are orthogonal, by construction. To unitize these we also note that they have the same magnitude, equal to $|H_{\mathbf{x}}| = \sqrt{p_x^2 + p_y^2 + 1/r^4}$. The 2BP is simple enough so that we can, by inspection, develop the additional orthogonal basis and its momentum as

$$\mathbf{u}_2 = [p_y, -p_x, y/r^3, -x/r^3]^T$$

$$\mathbf{v}_2 = -J\mathbf{u}_2 = [-y/r^3, x/r^3, p_y, -p_x]^T$$

These again all have the same magnitude, allowing them all to be unitized by a common term. By inspection we also note that these are all orthogonal to each other and to $\mathbf{u}_1$ and $\mathbf{v}_1$. These are all combined into the transformation matrix

$$R = \frac{1}{|H_{\mathbf{x}}|} \begin{bmatrix} p_x & p_y & x/r^3 & -y/r^3 \\ p_y & -p_x & y/r^3 & x/r^3 \\ -x/r^3 & y/r^3 & p_x & p_y \\ -y/r^3 & -x/r^3 & p_y & -p_x \end{bmatrix}$$

It is possible to show that this matrix is symplectic and is orthonormal.

We now provide an example computation of how this new coordinate system can be used to decompose the STM. Consider the orbit defined by a periapsis and apoapsis of 1 and 3, starting along the x axis. This has initial conditions $q_x = 1, q_y = 0, p_x = 0, p_y = \sqrt{3/2}$. We propagate this orbit over one half-period, to the state $q_x = -3, q_y = 0, p_x = 0, p_y = -1/\sqrt{6}$. The eigenvalues of this STM are computed at each time step and shown in Fig. 1. Note that the eigenvalues are quite dynamic, even though the complete orbit has only unity eigenvalues. This transient behavior of the STM eigenvalues is thought to have influence on the stretching behavior in the vicinity of the trajectory, as discussed and analyzed in [2]. One of the key uses of the analysis shown here is to help deconstruct the transient behavior that the STM experiences.

For the given initial conditions we have computed the STM Φ and the reduced Ψ matrix at times of 1/4 and 1/2 the period, to show as an example. For the 1/4 period mapping the associated sub-matrices are:

$$\Psi_{13} = \begin{bmatrix} 0.3740 & -7.4683 \\ 0 & 2.6736 \end{bmatrix}$$

with eigenvalues equal to the diagonal terms, where we note that they are reciprocals of each other. The monodromy matrix is then

$$\Psi_{24} = \begin{bmatrix} 0.0371 & 3.6400 \\ -0.2796 & -0.4777 \end{bmatrix}$$

with eigenvalues lying on the unit circle at an angle of $\pm 102.727°$.

For the 1/2 period mapping the sub-matrices are:

$$\Psi_{13} = \begin{bmatrix} 0.2676 & -35.6663 \\ 0 & 3.7370 \end{bmatrix}$$

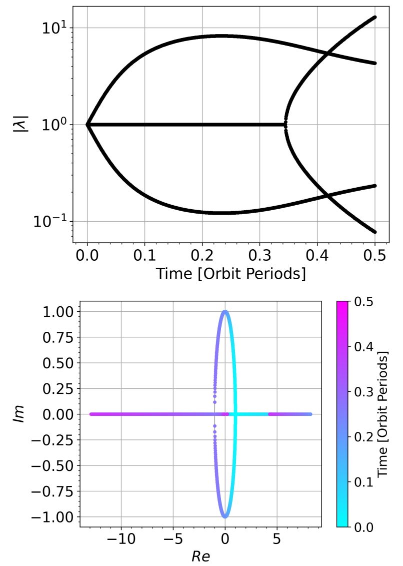

Figure 1: Eigenvalues of the STM evaluated over the first 1/2 orbit of the nominal trajectory. Note that if continued to the full period the eigenvalues would all collapse back to unity.

again with eigenvalues equal to the diagonal terms and reciprocal. The monodromy matrix is then

$$\Psi_{24} = \begin{bmatrix} -0.8028 & 0 \\ 0 & -1.2457 \end{bmatrix}$$

Now the monodromy matrix is unstable with eigenvalues on the negative real axis and reciprocal.

6 Conclusions

In this note the ortho-symplectic transformation evaluated along a trajectory is presented and described. The use of these transformations to derive a monodromy matrix map between any two points along a trajectory is defined, with examples from the 2BP. This monodromy map reduces to the usual monodromy matrix if evaluated over one period, but does not require a surface of section to be specified and uses the flow direction dynamics to define the surface. Applications of this mapping will be discussed at the meeting.

References

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[2] F.Y. Hsiao and D.J. Scheeres. 2006. "Evolution of Eigenvalues during the Transient Period for Hamiltonian Systems," Physica D: Nonlinear Phenomena 213: 66-75.